Ex 7: Integrating a Zoho Creator Application with Another Zoho Service using Connections and APIs

Topic: Patient Appointment Booking System

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1. Aim

To connect the Patient Appointment Booking application built in Zoho Creator with Zoho CRM so that a patient's booking details can be pushed into the CRM's Leads module as a new record, demonstrating how Zoho Creator's Connections framework and REST API calls are used to exchange data between two Zoho services.

2. Procedure

2.1 Problem Statement & Integration Scope

Bookings captured through the Patient_Appointment_booking form stay inside Zoho Creator, while the clinic's front-office and follow-up team would prefer to track prospective patients in Zoho CRM. Copying details from one system to the other by hand is slow and error-prone. This experiment removes that step by letting staff send a booking to CRM directly from the report with a single click.

- **Source application:** Patient Appointment Booking (Zoho Creator)
- **Target service:** Zoho CRM (Leads module)
- **Trigger:** A custom action (“Push to CRM”) on the Patient Appointment booking Report
- **Data transferred:** Patient name, Email, Phone, and the doctor / date / time slot of the booking

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2.2 Target Service Setup

A Zoho CRM account (CRM Teamspace) was opened with the same Zoho login that is used for Creator. Its standard modules — Leads, Contacts, Accounts and Deals — became available, and the Leads module serves as the destination for the integration.

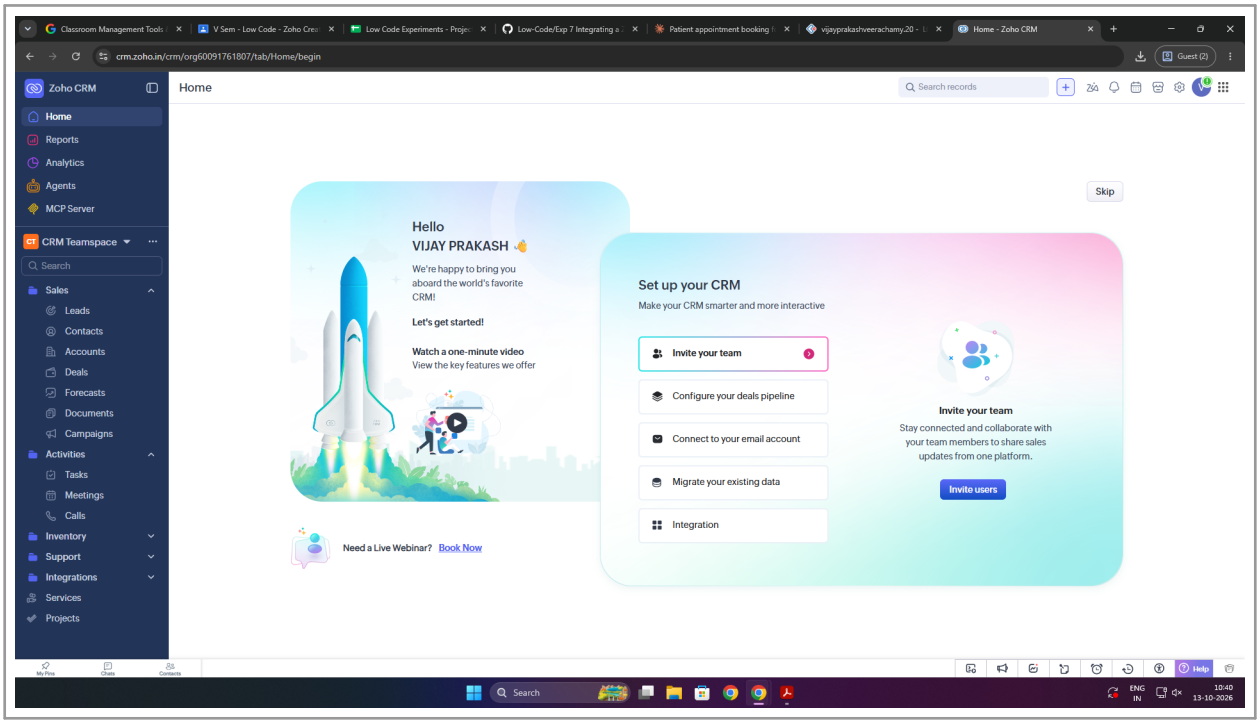


Figure 1: Zoho CRM Home Dashboard after Account Setup

2.3 Configuring the Native Connection

Zoho Creator manages authorisation to other Zoho services through its Connections feature, which handles the OAuth process without exposing tokens. It is reached from the account-level Microservices area of Creator Home rather than from an individual application's Settings. A Zoho CRM connection was set up with these values:

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- **Connection Name:** CRM_Connection
- **Version:** Zoho CRM V1
- **Authorization Access:** Administrators only
- **Connection Mode:** Quick connect (Zoho credential)
- **Access:** All Actions across the CRM modules

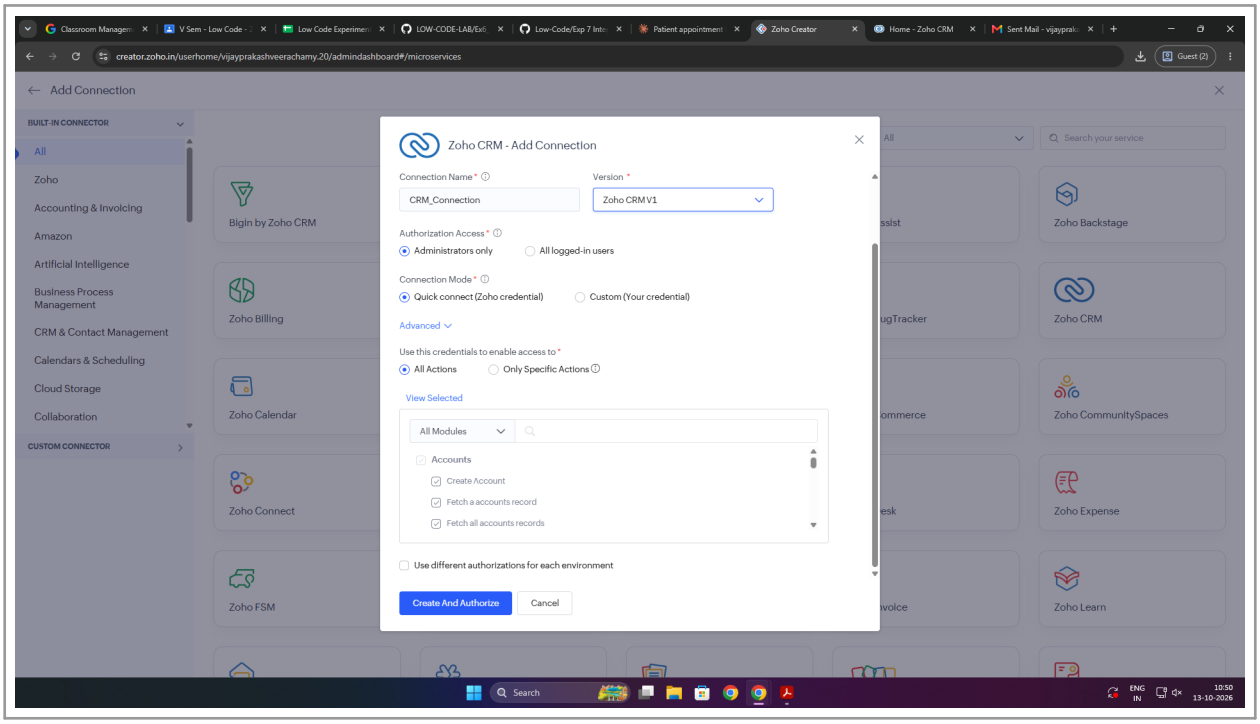


Figure 2: Zoho CRM – Add Connection Configuration

2.4 Limitation Encountered: Plan Restriction

Clicking “Create And Authorize” returned an “Upgrade connection count” message. Since this was the first connection attempted on the account, the message points to a plan-level limit on the expired trial rather than a mistake in the configuration.

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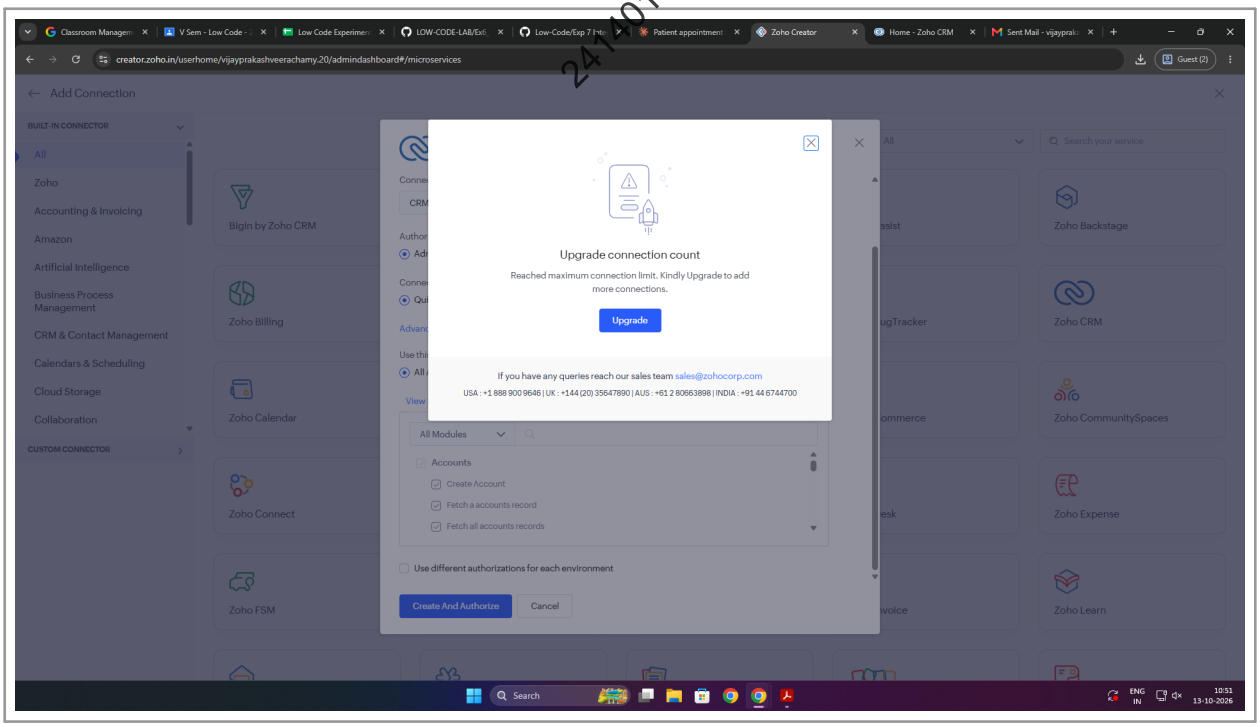
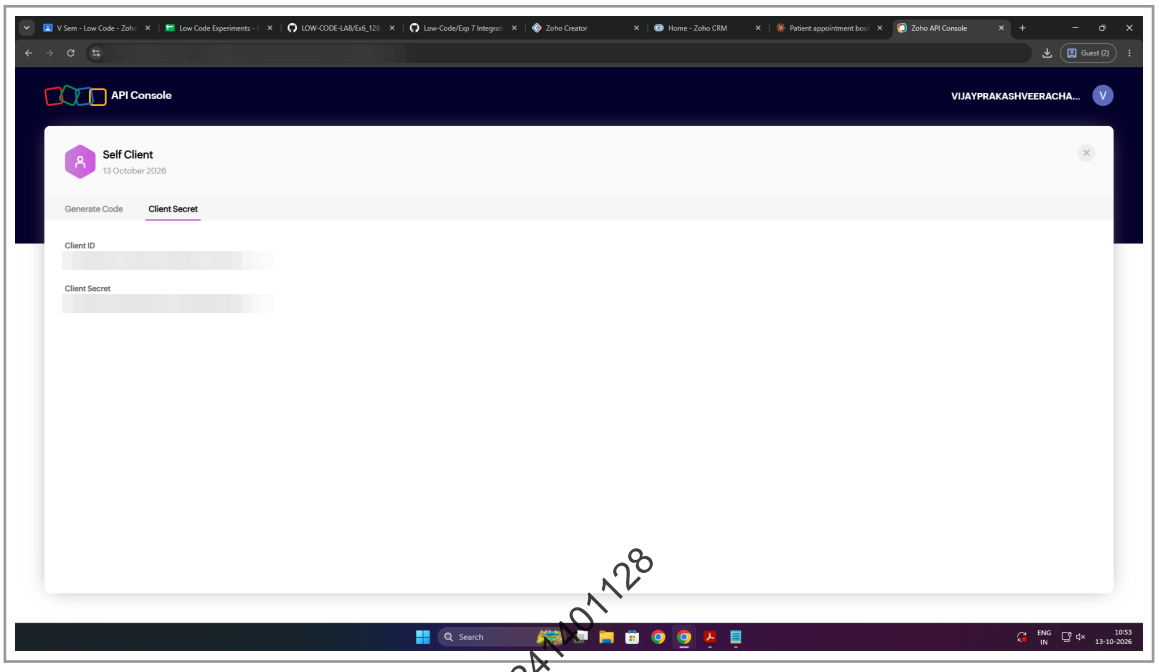


Figure 3: “Upgrade Connection Count” Message Returned by Zoho Creator

2.5 Alternative Approach: OAuth Self Client with invokeurl

To still meet the aim of integrating through connections and APIs, a manual OAuth 2.0 flow was used instead: a Self Client from the Zoho API Console provides the credentials, and Deluge's invokeurl task calls the Zoho CRM REST API directly.

1. A Self Client was created in the Zoho API Console (api-console.zoho.in), which generated a Client ID and Client Secret.
2. A grant code was generated from the Generate Code tab using the scope ZohoCRM.modules.ALL with a 10-minute validity.
3. The grant code was exchanged for an access token with a POST request made through the curl utility, since the token endpoint accepts only POST.
4. The returned access_token was placed in the Authorization header of a Deluge script attached to a custom action on the report.



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Figure 4: Self Client in the Zoho API Console (credentials masked)

```
curl.exe -X POST "https://accounts.zoho.in/oauth/v2/token" ^
  -d "grant_type=authorization_code" ^
  -d "client_id=<CLIENT_ID>" ^
  -d "client_secret=<CLIENT_SECRET>" ^
  -d "code=<GRANT_CODE>"
```

2.6 Deluge Script: Push Patient to Zoho CRM as a Lead

The script below was added as a custom action (“Push to CRM”) under Report → Quick View → Actions → Add New Action Item, set to run “At once” without a confirmation prompt, so it appears in the record-level options menu of the report.

```
lastName = ifnull(input.patient_name.last_name,"");
if(lastName == "")
{
    lastName = input.patient_name.first_name;
}
leadMap = Map();
leadMap.put("First_Name", input.patient_name.first_name);
leadMap.put("Last_Name", lastName);
leadMap.put("Email", input.Email);
leadMap.put("Phone", input.Phone);
leadMap.put("Lead_Source", "Zoho Creator - Patient Appointment");
leadMap.put("Description", "Doctor: " + input.Doctor + ", Date: "
    + input.Appointment_Date1 + ", Slot: " + input.time_slot);

headerMap = Map();
headerMap.put("Authorization", "Zoho-oauthtoken <ACCESS_TOKEN>");
dataList = List();
dataList.add(leadMap);
paramMap = Map();
paramMap.put("data", dataList);

response = invokeurl
[
    url: "https://www.zohoapis.in/crm/v2/Leads"
    type: POST
    parameters: paramMap.toString()
    headers: headerMap
];
info response;
```

Table 1 shows how the form fields were mapped to the CRM Leads fields.

Creator Field	CRM Lead Field	Purpose
patient_name (first name)	First_Name	Patient's first name
patient_name (last name)	Last_Name	Mandatory CRM identifier field
Email	Email	Patient contact email
Phone	Phone	Patient contact number
Doctor, Appointment_Date, Time_Slot	Description	Booking details as a note
(static value)	Lead_Source	Marks the record's origin as Zoho Creator

Table 1: Field Mapping between the Booking Form and Zoho CRM Leads

3. Output

Zoho CRM was set up and made ready as the target service (Figure 1). The native Zoho CRM connection was configured in Zoho Creator (Figure 2), but authorising it returned the plan-limit message shown in Figure 3. As an alternative, a Self Client was created in the Zoho API Console to obtain OAuth credentials (Figure 4), and the Deluge script in Section 2.6 was prepared to call the Zoho CRM Leads API through invokeurl.

4. Result

The Patient Appointment Booking application was prepared for integration with Zoho CRM using Zoho's connection and API framework. The native Connections feature could not be authorised because of a plan restriction on the trial account, so an OAuth 2.0 Self Client together with Deluge's invokeurl task was configured as the alternative route for pushing a booking to the CRM Leads module.